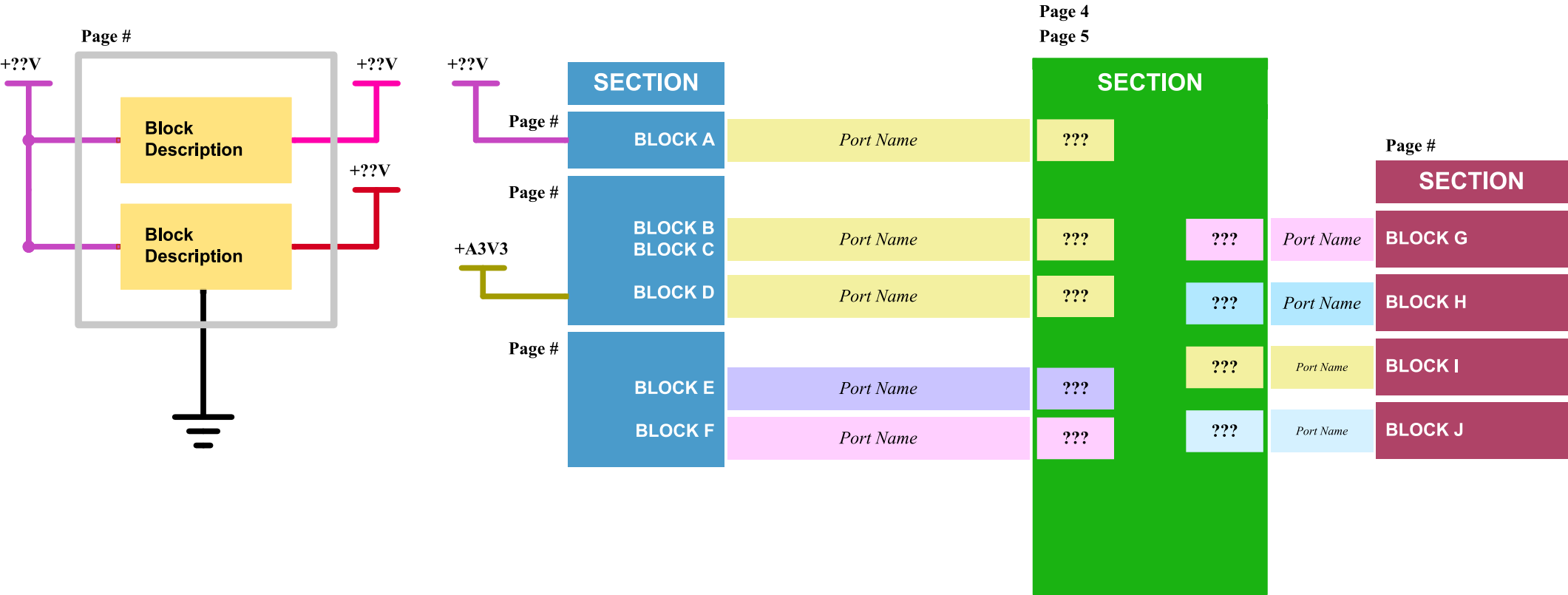


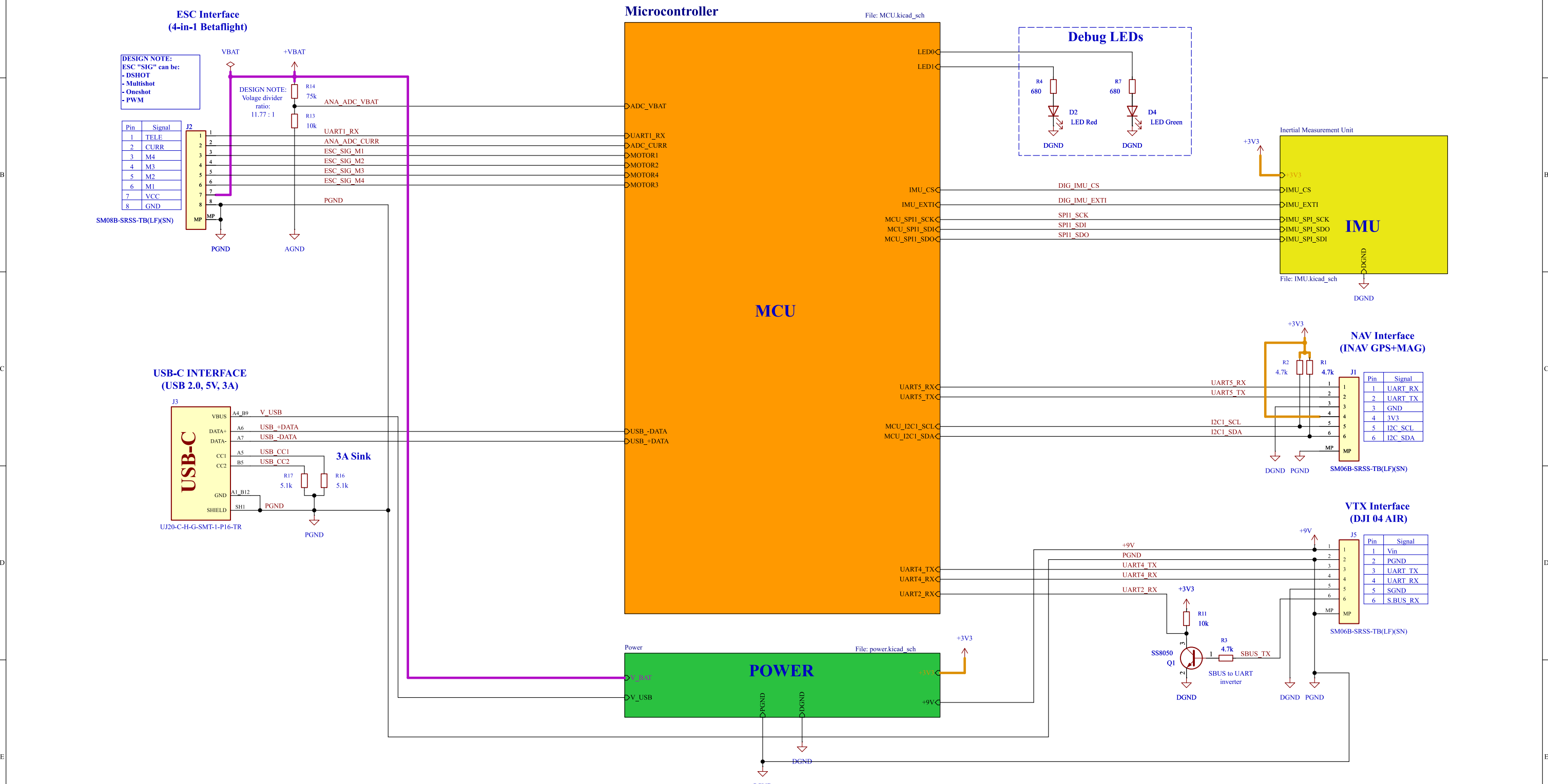
[2] Block Diagram



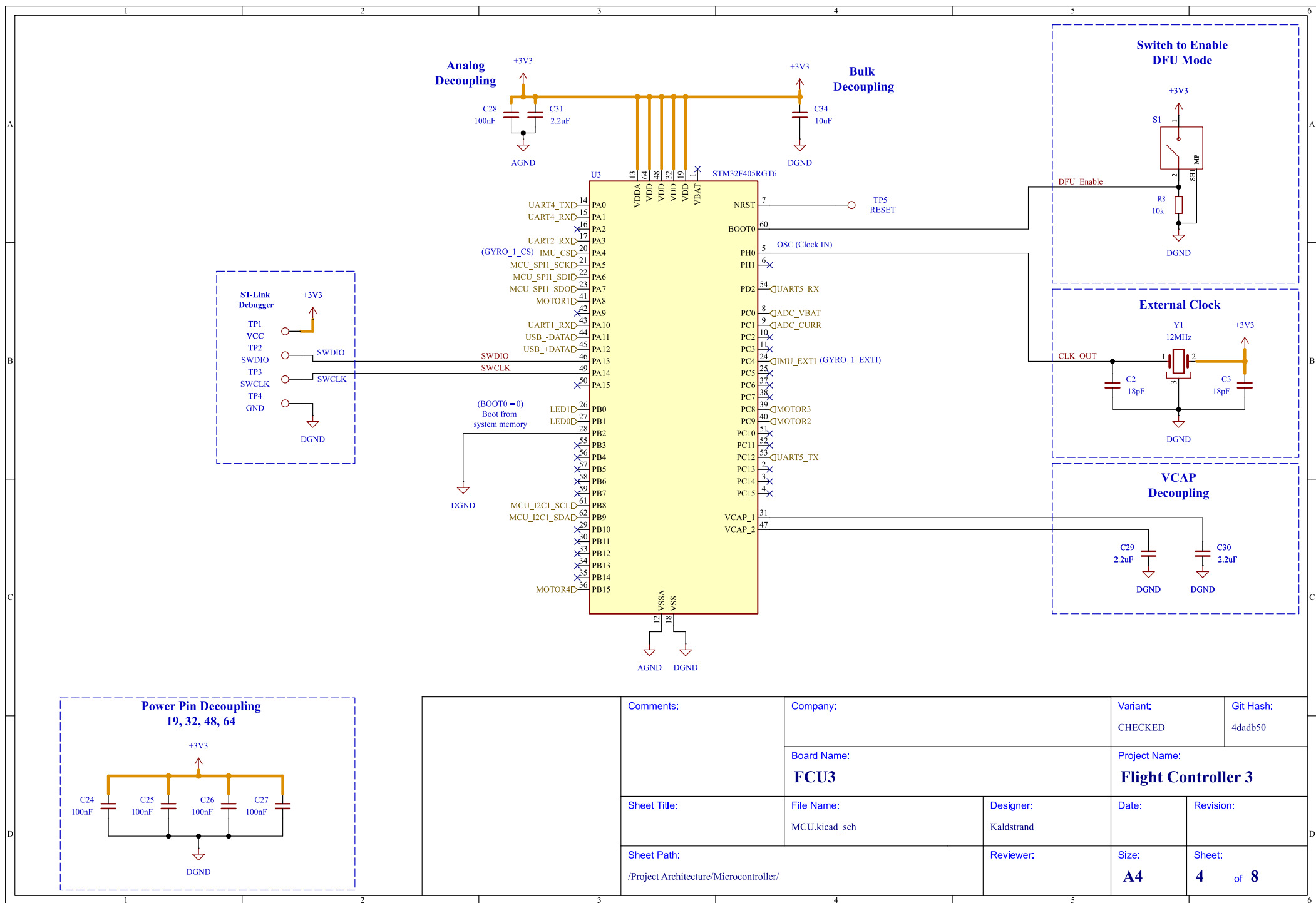
Target specifications:	
Input voltage:	?? - ?? V
Spec 2	??
Spec 3	??
Spec 4	??

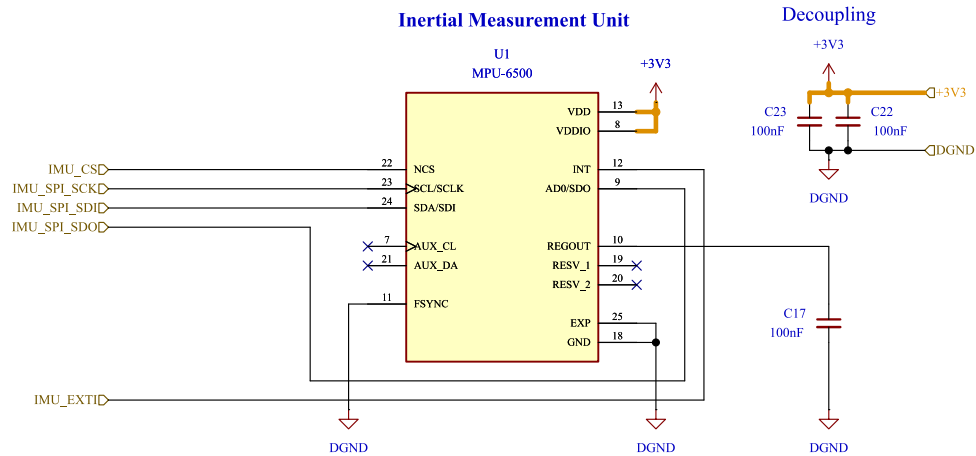
	Comments:		Company: Kaldstrand Electronics		Variant: CHECKED	Git Hash: 4dad50
			Board Name: FCU3		Project Name: Flight Controller 3	
	Sheet Title: Block Diagram	File Name: Block Diagram.kicad_sch	Designer: Kaldstrand	Date: 2025-01-12	Revision: + (Unreleased)	
	Sheet Path: /Block Diagram/		Reviewer:	Size: A3	Sheet: 2 of 8	

[3] Project Architecture

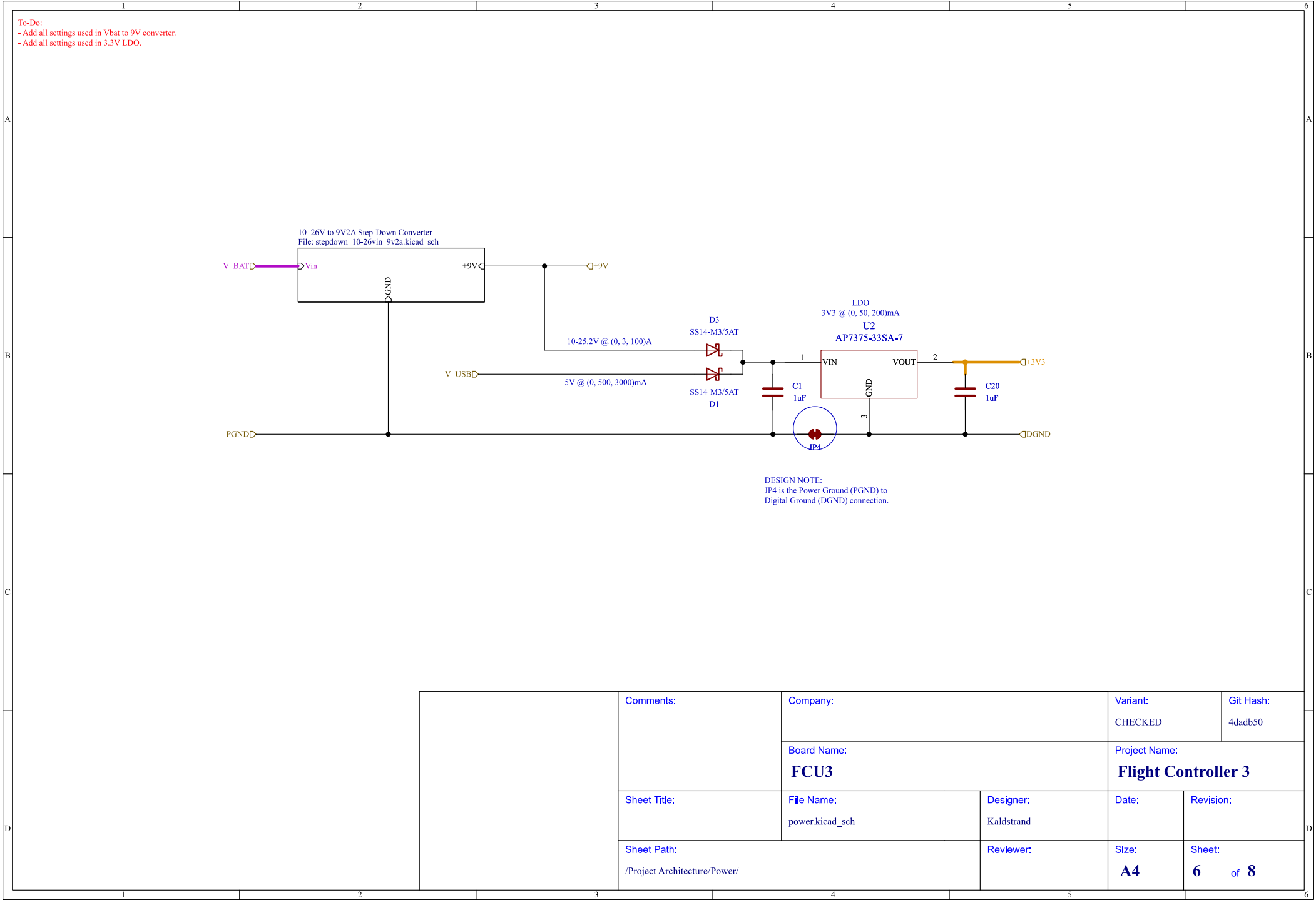


Grounding Strategy Ground Ties DGND JP3 AGND DGND AGND Mounting Holes H1 H2 H3 H4	PGND			Comments:		Company: Kaldstrand Electronics		Variant: CHECKED		Git Hash: 4dad50	
				Board Name: FCU3		Project Name: Flight Controller 3					
	Sheet Title: Project Architecture			File Name: Project Architecture.kicad_sch		Designer: Kaldstrand		Date: 2025-01-12		Revision: + (Unreleased)	
Sheet Path: /Project Architecture/				Reviewer:		Size: A3		Sheet: 3 of 8			





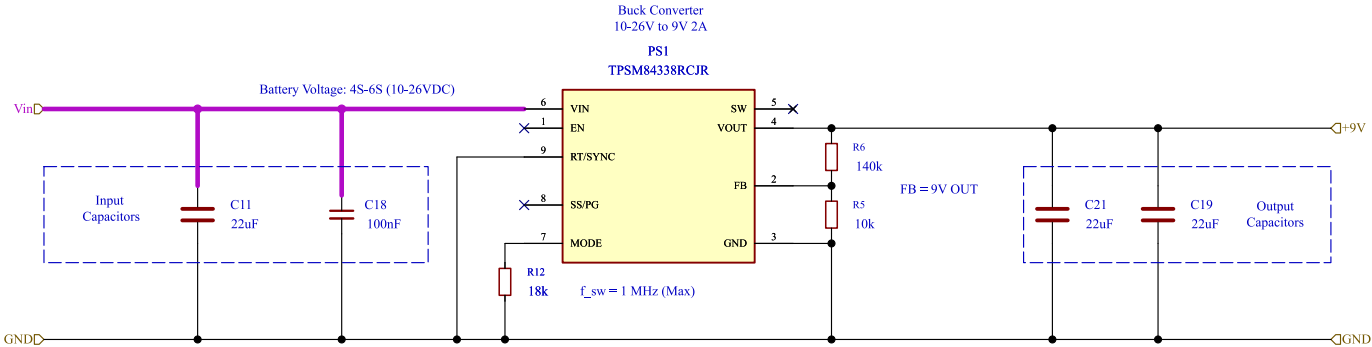
	Comments:	Company:		Variant: CHECKED	Git Hash: 4dad50
		Board Name: FCU3		Project Name: Flight Controller 3	
	Sheet Title:	File Name: IMU.kicad_sch	Designer: Kaldstrand	Date:	Revision:
	Sheet Path: /Project Architecture/Inertial Measurement Unit/		Reviewer:	Size: A4	Sheet: 5 of 8



DESIGN NOTE:
 $R_{FBB} = (V_{OUT} - V_{REF}) / V_{REF} * R_{FBB}$
 $R_{FBB} = (9V - 0.6V) / 0.6V * 10k$
 $R_{FBB} = 140k$

 $V_{OUT} = 9V, V_{REF} = 0.6V, R_{11} = 10k, R_6 = 140k$

To-Do:
Add design notes
explaining DC/DC settings.



	Comments:	Company:	Variant: CHECKED	Git Hash: 4dad50
		Board Name: FCU3	Project Name: Flight Controller 3	
	Sheet Title:	File Name: stepdown_10-26vin_9v2a.kicad_sch	Designer: Kaldstrand	Date: Revision:
	Sheet Path: /Project Architecture/Power/10-26V to 9V2A Step-Down Converter/	Reviewer:	Size: A4	Sheet: 8 of 8

[9] Revision History

	Comments:	Company: Kaldstrand Electronics		Variant: CHECKED	Git Hash: 4dad50
		Board Name: FCU3		Project Name: Flight Controller 3	
	Sheet Title: Revision History	File Name: Revision History.kicad_sch	Designer: Kaldstrand	Date: 2025-01-12	Revision: + (Unreleased)
	Sheet Path: /Revision History/		Reviewer:	Size: A4	Sheet: 9 of 8